DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

B.Tech. III Semester (MAIC-201)
Course Title :. Discrete Mathematics and statistical methods

Mid Term Examination-I
M.M.:15
Time: 6.00 p.m.-6.50 p.m.

NOTE: *Attempt all the Questions. Each question carries equal marks.*

Q1. State and prove Principle of inclusion and exclusion.

Q2. Fit a Binomial distribution for the following data and compare the theoretical frequencies with the actual ones:

X:	0	1	2	3	4	5
F:	2	14	20	34	22	8.

Q3. In a certain factory turning out razor blades, there is small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10, use Poisson distribution to calculate the approximate number of packets containing no defective, one defective, two defective blades respectively in a consignment of 10,000 packets.

Q4. The contents of three urns are : 1 white, 2 red, 3 green balls; 2 white, 1 red, 1 green ball and 4 white, 5 red, 3 green balls. Two balls are drawn from an urn chosen at random. These are found to be one white and one green. Find the probability that the balls so drawn come from the third urn.

Q5. Prove that the Kuratowski's graph K_5 is non planar.